

Chapter 2 Test

Dynamics

K	18/18	T	17/18	A	10/10	C	9/10
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SHOW ALL YOUR WORK IN ANSWERING QUESTIONS.

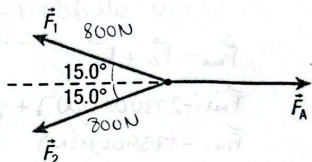
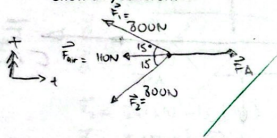
Well Done!

THINKING

1. A tow-plane (plane that tows a glider) is pulling two gliders (airplane which does not depend on engine to fly) at a constant velocity with a net force of zero. Each glider is attached to a rope behind the tow-plane at the same altitude, but 15.0° off the path of the tow-plane, as shown. The tension in the two ropes F_1 and F_2 is the same at 8.00×10^2 N. There is an air-resistance force of 1.10×10^3 N against the applied force along the path of the tow plane. Determine the force applied (F_A) by the propeller on the tow-plane.

Show all your work

(3 marks)



$$\vec{F}_A + \vec{F}_1 + \vec{F}_2 + \vec{F}_{air} = 0$$

$$\vec{F}_A = -\vec{F}_1 - \vec{F}_2 - \vec{F}_{air}$$

$$\vec{F}_{1x} = 800 \cos 15^\circ = 772.741 \text{ N [W]}$$

$$\vec{F}_{1y} = 800 \sin 15^\circ = 207.055 \text{ N [N]}$$

$$\vec{F}_{2x} = 800 \cos 15^\circ = 772.741 \text{ N [W]}$$

$$\vec{F}_{2y} = 800 \sin 15^\circ = 207.055 \text{ N [S]}$$

$$\vec{F}_{air} = 1100 \text{ N [W]}$$

$$\vec{F}_{netx} = \vec{F}_{1x} + \vec{F}_{2x} + \vec{F}_{air} + \vec{F}_A$$

$$0 = 772.741 \text{ N [W]} + 772.741 \text{ N [W]} + 1100 \text{ N [W]} + \vec{F}_A$$

$$\vec{F}_A = -1655.482 \text{ N [E]}$$

$$\vec{F}_{nety} = \vec{F}_{1y} + \vec{F}_{2y} + \vec{F}_{Ay}$$

$$0 = 207.055 \text{ N [N]} + 207.055 \text{ N [S]} + \vec{F}_{Ay}$$

$$\vec{F}_A = 0 \text{ N}$$

$$\therefore \vec{F}_A = 1655.482 \text{ N [E]}$$

$$\text{Final answer: } \vec{F}_A = 1655.48 \text{ N [E]}$$

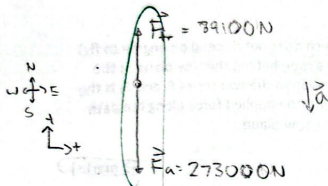
2. A 40.0 ton (3.63×10^4 kg) luxury yacht cruising in the sea, resulting in a force of 2.73×10^5 N [South] on the yacht. The force of all frictions at this point on the yacht is 2.91×10^5 N [North].

- a) Draw a well labelled free-body diagram (FBD) for the yacht.
b) What is the acceleration of the yacht?

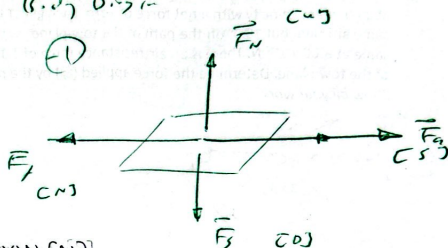
Show all your work

3 model

[4 marks]



incompl. 1. Friction
B.O.D. Diagram



$$\vec{F}_{\text{net}} = \vec{F}_a + \vec{F}_f$$

$$\vec{F}_{\text{net}} = -273000 \text{ N} + 291000 \text{ N}$$

$$\vec{F}_{\text{net}} = 18000 \text{ N}$$

$$\vec{F}_{\text{net}} = m\vec{a}$$

$$\vec{a} = \frac{\vec{F}_{\text{net}}}{m}$$

$$\vec{a} = \frac{18000 \text{ N}}{36300 \text{ kg}}$$

$$\vec{a} = 0.066 \text{ m/s}^2 \text{ [N]}$$

Final answer: $\vec{a} = 0.066 \text{ m/s}^2 \text{ [N]}$

3. An electron is a subatomic particle that is subject to electric forces. An electron moving in the +x direction accelerates from an initial velocity of 5.40×10^5 m/s to a final velocity of 2.10×10^6 m/s while traveling a distance of 0.038 m. The electron's acceleration is due to two electric forces parallel to the x axis: $F_1 = 1.75 \times 10^{-16}$ N, and F_2 , which points in the -x direction. Note that electron mass is 9.11×10^{-31} kg.

- a) Determine the magnitude of the net force acting on the electron.
b) Calculate the magnitude of the electric force, F_2 .

Show all your work

a) $F_{net} = m\vec{a}$

$m = 9.11 \times 10^{-31}$ kg

$\vec{a} = ?$

$\vec{a} = \frac{v_2^2 - v_1^2}{2d}$

$v_2 = 2.1 \times 10^6$ m/s

$v_1 = 5.4 \times 10^5$ m/s

$d = 0.038$ m

$\vec{a} = \frac{(2.1 \times 10^6)^2 - (5.4 \times 10^5)^2}{2 \cdot 0.038}$

$\vec{a} = 5.418947 \times 10^{13}$ m/s²

$F_{net} = m\vec{a}$

$F_{net} = 9.11 \times 10^{-31} \cdot 5.418947 \times 10^{13}$

$\therefore F_{net} = 4.93666 \times 10^{-17}$ N

b) $F_{net} = F_1 + F_2$

$F_2 = F_{net} - F_1$

$F_2 = 4.93666 \times 10^{-17}$ N - 1.75×10^{-16} N

$\therefore F_2 = -1.256 \times 10^{-16}$ N

(4 marks)

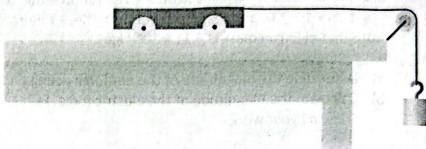
Final answer: a) $F_{net} = 4.937 \times 10^{-17}$ N

Final answer: b) $F_2 = -1.256 \times 10^{-16}$ N

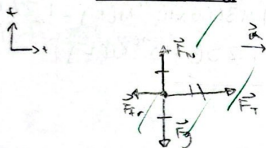
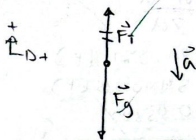
No
Need
for this

4. A string is tied to 1.9 kg cart on a table and 1.2 kg object hanging over a pulley as shown. The coefficient of kinetic friction between the cart and the table is 0.2.

- Draw a free body diagram (FBD) of the cart and another free body diagram (FBD) of the object.
- Calculate the magnitude of the acceleration of the cart and the hanging object.
- Calculate the magnitude of the tension in the string.
Show all your work



[7 marks]

FBD of cart (1.9 kg)FBD of object (1.2 kg)

Cart

$$\begin{aligned}
 \vec{F}_{\text{net}} &= m\vec{a} \\
 \vec{F}_{\text{net}} &= \vec{T} - \vec{F}_f \\
 m\vec{a} &= \vec{T} - \vec{F}_f \\
 m_1\vec{a} + \vec{F}_f &= \vec{T}
 \end{aligned}$$

obj

$$\begin{aligned}
 \vec{F}_{\text{net}} &= m\vec{a} \\
 \vec{F}_{\text{net}} &= \vec{F}_g - \vec{T} \\
 m\vec{a} &= \vec{F}_g - \vec{T} \\
 \vec{T} &= \vec{F}_g - m\vec{a}
 \end{aligned}$$

Cart moving 2.592 m/s^2 [E]

Object moving 2.592 m/s^2 [D]

c) $m_2\vec{a} = \vec{F}_g - \vec{T}$

$\vec{T} = \vec{F}_g - m_2\vec{a}$

$\vec{T} = m_2\vec{g} - m_2\vec{a}$

$\vec{T} = (1.2 \cdot 9.8) - (1.2 \cdot 2.592)$

$\vec{T} = 8.65 \text{ N [u]}$

$m_1\vec{a} + \vec{F}_f = \vec{F}_g - m_2\vec{a}$

$m_1\vec{a} + m_2\vec{a} = \vec{F}_g - \vec{F}_f$

$\vec{a}(m_1 + m_2) = \vec{F}_g - \vec{F}_f$

$\vec{a} = \frac{\vec{F}_g - \vec{F}_f}{(m_1 + m_2)}$

$\vec{F}_g = m_2\vec{g}$

$\vec{F}_f = \mu_k \cdot \vec{F}_N$

$\vec{F}_N = \vec{F}_g = m_1\vec{g}$

$\vec{F}_f = (m_1\vec{g}) \cdot \mu_k$

$\vec{a} = \frac{m_2\vec{g} - (m_1\vec{g}) \cdot \mu_k}{(m_1 + m_2)}$

$\vec{a} = \frac{1.2 \cdot 9.8 - (1.9 \cdot 9.8) \cdot 0.2}{(1.9 + 1.2)}$

$\vec{a} = 2.592 \text{ m/s}^2$ [CW]

Final answer: b) $\vec{a} = 2.592 \text{ m/s}^2$

Final answer: c) $\vec{T} = 8.65 \text{ N}$

Chapter 2 Test

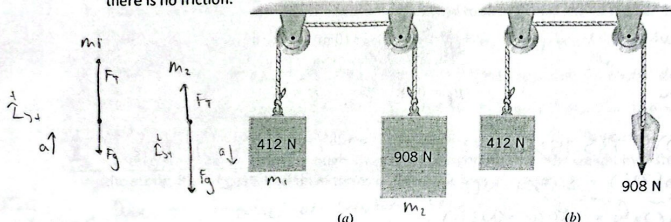
Dynamics

K	T	A	C
/18	/18	10/10	9/10

SHOW ALL YOUR WORK IN ANSWERING QUESTIONS.

APPLICATION

1. Two blocks are connected by a rope that passes over a set of pulleys, as part (a) of the drawing shows. One block has a weight of 412 N, and the other has a weight of 908 N. The rope and the pulleys are massless and there is no friction.



[5 marks]

- What is the acceleration of the lighter block?
- Suppose that the heavier block is removed, and a downward force of 908 N is provided by someone pulling on the rope, as part (b) of the drawing shows. Determine the acceleration of the remaining block.
- Explain why the answers in (a) and (b) are different.

Show all your work

$$a) \quad 412 \text{ N} = 41.3 \text{ kg} \quad 908 \text{ N} = 92.559 \text{ kg}$$

$$m_1: F_{\text{net}} = F_T - F_g \quad m_2: F_{\text{net}} = F_T - F_g$$

$$F_{\text{net}} = ma \quad m_2 a = F_T - F_g$$

$$F_T = m_1 a + F_g \quad F_T = F_g + m_2 a$$

$$m_1 a + F_g = F_g + m_2 a$$

$$m_1 a + m_2 a = F_g - F_g$$

$$a = \frac{F_g - F_g}{(m_1 + m_2)}$$

$$a = \frac{412 - 908}{(41.3 + 92.559)}$$

$$a = -3.69 \text{ m/s}^2$$

$$\therefore \vec{a} = 3.69 \text{ m/s}^2 \text{ [U]}$$

$$b) \quad \begin{aligned} F_T &= 908 \text{ N} \\ F_g &= 412 \text{ N} \\ 412 \text{ N} &= 41.3 \text{ kg} \end{aligned}$$

$$F_{\text{net}} = ma$$

$$F_{\text{net}} = F_T - F_g$$

$$a = \frac{F_T - F_g}{m}$$

$$a = \frac{908 - 412}{41.3}$$

$$\therefore \vec{a} = 11.81 \text{ m/s}^2 \text{ [U]}$$

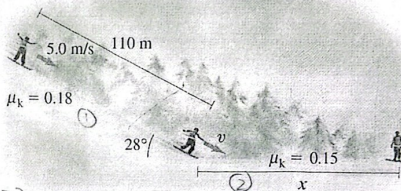
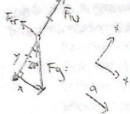
- c) The accelerations are different because in the first example the 908 N weight is acting, while the second scenario only has one mass with the same net force acting.

Final answer: a) $3.69 \text{ m/s}^2 \text{ [U]}$ Final answer: b) $11.81 \text{ m/s}^2 \text{ [U]}$

Final answer: c) two masses vs. one mass

2. A 75 kg snowboarder has an initial velocity of 5.0 m/s at the top of a 28° incline, as shown in the diagram. After sliding down the 110 m long incline (on which the coefficient of kinetic friction is 0.18 on that sloped surface), the snowboarder has attained a velocity v . The snowboarder then slides along a flat surface (on which the coefficient of kinetic friction is 0.15 on the flat levelled surface) and comes to rest after a distance x . Use Newton's second law to determine the snowboarder's acceleration while on the incline and while on the flat surface until stopping. Then use these accelerations to determine distance x , as shown in the diagram. Show all your work

[5 marks]



① $75 \text{ kg} \cdot 9.81 \text{ m/s}^2 [\text{D}] = 735.75 \text{ N} [\text{D}]$
 $F_{gx} = 735.75 \text{ N} \sin(28^\circ) = 345.41 \text{ N} [\text{F}]$
 $F_{gy} = 735.75 \text{ N} \cos(28^\circ) = 649.63 \text{ N} [\text{D}]$

$F_{fr} = F_N \cdot \mu_k = F_{gy} \cdot \mu_k$
 $F_{fr} = 649.63 \text{ N} \cdot 0.18 = 116.93 \text{ N} [\text{B}]$

$F_{\text{net}} = F_{gx} - F_{fr}$
 $F_{\text{net}} = 345.41 \text{ N} [\text{F}] + (-116.93 \text{ N} [\text{B}])$
 $F_{\text{net}} = 228.48 \text{ N} [\text{F}]$

$F = ma$
 $a = \frac{F_{\text{net}}}{m}$
 $a = \frac{228.48}{75 \text{ kg}}$
 $a = 3.05 \text{ m/s}^2 [\text{F}]$

$v_i = 5 \text{ m/s} [\text{F}]$
 $v_f = ? \text{ m/s}$
 $a = 3.05 \text{ m/s}^2 [\text{F}]$
 $\Delta d = 110 \text{ m}$

$v_f^2 = v_i^2 + 2a\Delta d$

$v_f = \sqrt{v_i^2 + 2a\Delta d}$

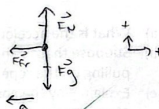
$v_f = \sqrt{5^2 + 2(3.05)(110)}$

$v_f = 26.38 \text{ m/s} [\text{F}]$

$v_i = 26.38 \text{ m/s} [\text{F}]$
 $v_f = 0 \text{ m/s} [\text{F}]$
 $a = 1.4715 \text{ m/s}^2 [\text{B}]$
 $\Delta d = ?$

$v_i^2 = v_f^2 + 2a\Delta d$
 $a = \frac{v_f^2 - v_i^2}{2\Delta d}$
 $\Delta d = \frac{0^2 - 26.38^2}{2(-1.4715)}$

$\Delta d = 236.49 \text{ m} [\text{F}]$



② $F_{fr} = F_N \cdot \mu_k = F_{gy} \cdot \mu_k$
 $F_{fr} = m \cdot g \cdot \mu_k = 75 \text{ kg} \cdot 9.81 \cdot 0.15$
 $F_{fr} = 110.36 \text{ N} [\text{B}]$
 $F_{\text{net}} = F_{fr} + F_{gx}$
 $F_{\text{net}} = 110.36 \text{ N} [\text{B}]$
 $a = \frac{F_{\text{net}}}{m}$
 $a = \frac{110.36}{75}$
 $a = 1.4715 \text{ m/s}^2 [\text{B}]$

Final answer: $\Delta d = 240 \text{ m} [\text{F}]$

N. need
for k4.s
Page 2

Name: _____

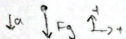
Date: _____

COMMUNICATION

1. A ball thrown straight up has zero velocity at its highest point. Is the ball in equilibrium at this point? Why or why not?

The ball is not in equilibrium, as the net force isn't zero because gravity is still acting on it. Even though it is not moving, gravity's acceleration is still acting on it.

[2 marks]



2. When a car makes a sharp turn, the passengers tend to slide to one side of the car. Explain why this happens.

Due to Newton's first law of inertia, people slide when sharp turns are made. As the car accelerates forward, the people do as well, and when the car applies a force to make a sharp turn, the bodies still have energy or inertia forwards, and not sideways until the turn occurs.

[2 marks]

3. When a string barely strong enough lifts a heavy weight, it can lift the weight by a steady pull; but if you jerk the string, it will break. Explain in terms of Newton's laws of motion.

See solutions! Due to impulse or Newton's second law, a string will break. $F = ma$ states that force & acceleration are proportional, meaning the force applied on the string will be higher if pulled fast. Inertia of the object also makes rapid change of acceleration require a lot of force. $\vec{F} = m\vec{a}$ $\vec{a} = \frac{\Delta \vec{v}}{\Delta t}$ Δt is small, so $\vec{a}$ is large.

[2 marks]

4. It is often said that "friction always opposes motion." Give at least one example in which:

- a) Static friction causes motion.

Walking or running requires grip along a surface, and static friction allows shoes to grip on surfaces, therefore allowing motion of the whole body.

- b) Kinetic friction causes motion

The tires of a car require friction to move the car along the surface, and although it may slip, the kinetic friction still allows for the car's motion. How?

See solutions!

[2 marks]

Communication marks are awarded throughout the test according to the shown rubric.

[2 marks]

	Level 1	Level 2	Level 3	Level 4
Correct use of mathematical/physical symbols, labels, and conventions (e.g. units, proper component form for the values, positive direction given, etc...)	Few mathematical/physical conventions are used correctly.	Some mathematical/physical conventions are used correctly.	Most mathematical/physical conventions are used correctly.	Consistently uses mathematical/physical conventions correctly.
	[1/2]	[1]	[1-1/2]	[2]